

Capitolo 8

Lesson 2-12-2025

8.1 Mathematical representation of OFDM, noise of a sub-carrier; terminology used by the book; pulse shaping for OFDM

Given that the symbols we're generating and want to transmit are $x[k]$, we have that after IDFT and cyclic prefix, the OFDM waves will be

$$\bar{x}[n] = \frac{1}{K} \sum_{k=0}^{K-1} x[k] e^{j \frac{2\pi}{K} nk} \quad n = -L, \dots, K-1 \quad (2.211)$$

■ Transmitted samples after the IDFT and cyclic prefix addition

$$\bar{x}[n] = \frac{1}{K} \sum_{k=0}^{K-1} x[k] e^{j \frac{2\pi}{K} nk} \quad n = -L, \dots, K-1$$

■ Received signal after discarding the cyclic prefix (first L samples)

$$\begin{aligned} y[n] &= \sqrt{G} \sum_{\ell=0}^L h[\ell] \bar{x}[n - \ell] \\ &= \sqrt{G} \sum_{\ell=0}^L h[\ell] x[(n - \ell)_K] \quad n = 0, \dots, K-1 \end{aligned} \quad \Rightarrow \quad \begin{aligned} &\text{(now including noise)} \\ y[k] &= \sqrt{G} h[k] x[k] + v[k] \end{aligned}$$

Looks like a circular convolution
between length K signals
(channel zero padded)

Simple equalization
✓ Noise remains IID and $\sim \mathcal{N}_c(0, N_0)$

Per-subcarrier noise

$$\frac{N_0}{G} \frac{1}{|h[k]|^2}$$

Professor never used the *circular convolution*. We sum the noise which is IID and Gaussian.

Note:

Vedi: ex 2.22, 2.23 (esercizi esame molto simili a quelli del libro + vedi moodle) We can see that each **subcarrier has a noise variance** of

$$\frac{N_0}{G} \frac{1}{|h[k]|^2}. \quad (2.221)$$

This follows from the type of equalization we're performing (specifically by using (2.219) to find the noise covariance by considering $\hat{x}[k]\hat{x}^*[k]$ for symbols $x_k \equiv x[k] = 0$, so to only consider the noise),

My explanation

At a certain point we get that (with reference to A signal processing perspective (pp. 57-130), p. 50):

$$y[n] = \sum_{k=0}^{K-1} x_k h[k] e^{j\frac{2\pi}{K}kn},$$

which is the IDFT of $x_k h[k]$! We know that a zero forcing equalizer W is such that in the frequency domain, we'll have

$$W^*(z) \cdot H(z) = z^{-\Delta}.$$

What we see here is that we have a *dummy frequency domain* which sees the multiplication of x_k and $h[k]$: this will allow us to say that if we choose

$$x_k = \frac{1}{Kh[k]} \text{DFT}[y[n] | k],$$

the relation will be satisfied with $\Delta = 0$: in this sense, what we previously found can be considered a ZF equalizer.

If we then were to also add the channel gain $\sqrt{G}$ into the mix and we defined $\hat{x}[k] \equiv x_k$ and $y[k] = \text{DFT}[y | k]$, we have that the previous equation can be written just like in the book,

$$\hat{x}[k] = \frac{y[k]}{\sqrt{G}h[k]} \quad k = 0, \dots, K-1. \quad (2.219)$$

If the channel is small for a certain k , the noise is increasing in the subcarriers. If the frequency response is small, we're paying this in increasing noise.

This is somehow tied to ~d45a40.

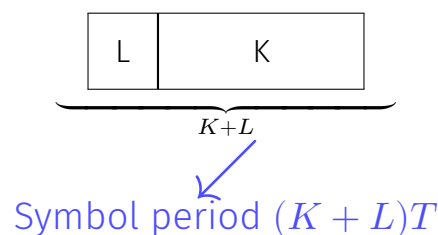
Terminology:

- Input to the IDFT are symbols
- Output of the IDFT are OFDM chips
- Interval between chips is T (would be the symbol period)
- Number of subcarriers is K , the dimension of the IDFT / DFT
- Collection of chips $\bar{x}[-L], \dots, \bar{x}[K-1]$ forms an OFDM symbol
- OFDM symbol period is $(K+L)T$
- Subcarrier spacing is the distance between adjacent subcarriers $\frac{1}{KT}$
- Effective pulse shape is usually a rectangular
- Edge subcarriers may be zero, leading to a spectral efficiency penalty

About chips: professor does not agree to this; *chip* has to do with slicing.

We have that $x[k]$ are symbols, $z[n]$ are chips.

The **interval between chips** is T . The **number of subcarriers** is K ; this is also the dimension of the IDFT/DFT. The collection of chips $\bar{x}[-L], \dots, \bar{x}[K-1]$ forms an **OFDM symbol**: an OFDM symbol is $K + L$.



It's tricky to have to do with this: we do S/P, then DFT, then we add the cyclic prefix. We're adding some symbols: we'll need to go faster in order to keep up with the previous frequency.

One way of choosing the pulse shape is to have a rectangular pulse of duration KT , giving that subcarriers will bring a signal of $KT \text{sinc}(fKT)$. In discrete frequency, the subcarriers are spaced of $1/K$, which, with a sampling period of T , gives that the K subcarriers will **span a total bandwidth of $1/T$** .

The actual (non-ideal) signal will surely have a bandwidth larger than $1/T$, forcing us to set some of the outer carriers to 0. This can be done either by *windowing*, aka by putting specific subcarriers to θ or by just applying a filter before transmission.

One thing which is typically done to improve the overall situation is to use raised cosines or similar signals.

$K \uparrow$ implies that the subcarriers are narrower and decay faster in frequencies. This term directly influences the containment of the signal in $1/T$. Also, the higher this value, the more the subcarriers are susceptible to little changes. (e.g.: if a subcarrier has a small phase shift, this causes more trouble if the symbols are tightly close). As it's clear, K also influences the resolution of the channel's frequency selectivity.

Comparisons

The book makes some comparisons between OFDM and single-carrier techniques. I just skipped over that. If interested, see *A signal processing perspective* (pp. 57-130), p. 53.

One last thing is that whenever the book (or this document) mentions *OFDM resource elements*, we're referring to a specific OFDM symbol in time and a specific subcarrier in frequency, according to the figure:

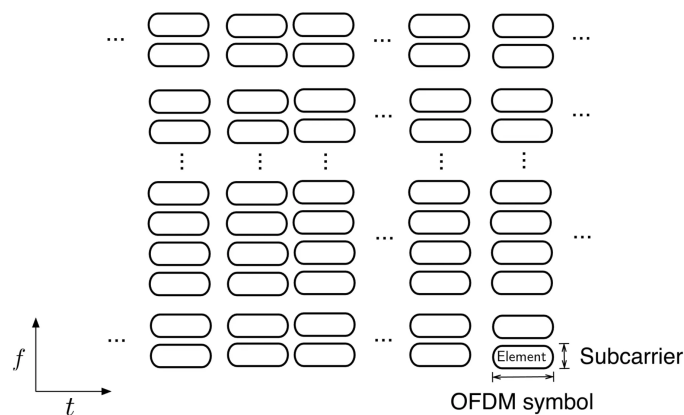


Figura 8.1: OFDM resource elements representation

8.2 Subcarrier spacing (+about the rectangular pulse shape of the signal), spectral efficiency penalty

Subcarrier spacing between adjacent carriers is $1/KT$; this comes from the writing

$$\exp\left(j\frac{2\pi}{K}kn\right) = \exp\left(j2\pi\frac{k}{KT}nT\right).$$

in which k/KT is the frequency term and nT is the time term. The spacing in frequency between each subcarrier will thus be $1/KT$.

The pulse shape is usually rectangular. We'll eventually need to go back to the continuous time domain; if we go back, we'll see that we have $x[n]$ (the transmission will happen in continuous time, of course):

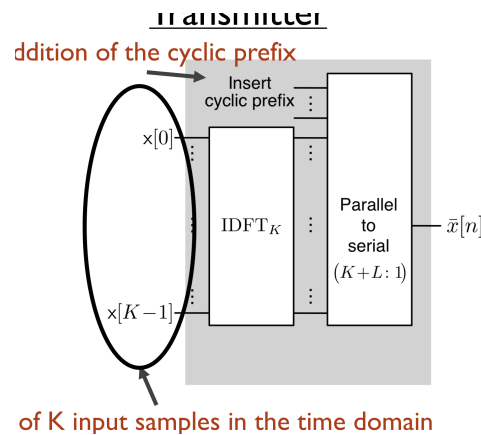


Figure 8.2: Transmitter: addition of the cyclic prefix

We'll actually need to use a rect:

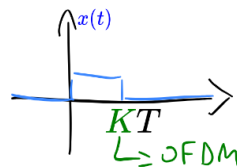


Figure 8.3: Rectangular pulse of duration KT

This is pretty brutal, but we'll have a sinc in the frequency domain; all the multiples of $1/KT$ will be 0:

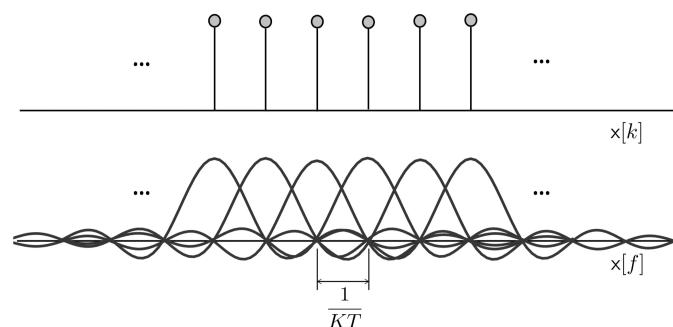


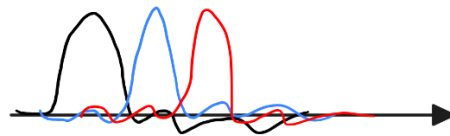
Figure 8.4: Sinc functions in frequency domain spaced by $1/KT$

This will allow for a perfect representation of our signal!

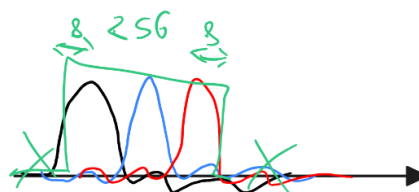
Between OFDM symbols we have KT seconds in time; if we consider them, the rect is KT and the sinc in frequency will go to zero on the multiples of $1/KT$: this means that the carriers have no influence over each other whatsoever.

Edge subcarriers might be 0 (we'll see why shortly), leading to a **spectral efficiency penalty**.

Let's say we have a certain number of subcarriers:



We know that if our filter is a rect, it might interfere, cutting away a portion of the signals in the frequency domain:



We'd like to filter beforehand, but we can't generally do this; what we thus do is to **switch off the edge carriers** (in the example up here we switch off the first and last 8 edge carriers). This means that the filter won't cut that sinc and it won't interfere with neighboring subcarriers.

In the following paragraph a more formal definition of penalty factor is given:

Penalty factor

A completely separate but equally important definition is that of **penalty factor**: it is given by how much the usable bandwidth is in relation to the used bandwidth; the point is that in reality we'll always have some excess bandwidth which will go over our constraints: just to give an example, in *example 2.23* the subcarrier spacing is 15KHz but it turns out that the actual subcarrier bandwidth is $\frac{500\text{MHz}}{300} = 16.66\text{KHz}$, more than what would be available. The penalty factor is thus calculated as

$$\text{Penalty factor} = \frac{\text{Usable bandwidth}}{\text{Used bandwidth}};$$

this value is useful as whenever we have a channel of known spectral efficiency R/B but we also know that we're exceeding the channel's bandwidth, we can correct the spectral efficiency as

$$(\text{Penalty factor}) \cdot \frac{R}{B}.$$

Just as an extra piece of information, another way to see this is

$$\text{Penalty factor} = \frac{\text{Theoretically used bandwidth}}{\text{Actually used bandwidth}}.$$

8.3 Examples of small exercises; many definitions of small values related to OFDM

In the example we turn off the first and last 8 subcarriers for a system for which we have $K = 256$: we'll have an **efficiency** which is limited:

$$\frac{240}{256} < 1.$$

We're thus having a total efficiency of

$$\frac{256}{L + 256} \cdot \frac{240}{256} < 1.$$

We have some examples which **can also be conceived as some exam exercises** (as can be seen, they were indeed asked in the multiple-choices part of the exam).

Example 2.22

Let $T_{\text{active}} = 3.2 \mu\text{s}$ with $K = 256$ subcarriers and a cyclic prefix of length $L = 64$. Find the chip period T , the passband bandwidth $1/T$, the subcarrier spacing, and the guard interval spanned by the cyclic prefix.

Solution

The chip period T satisfies $(256)T = 3.2 \mu\text{s}$ and thus $T = 12.5 \text{ ns}$. Then, $1/T = 80 \text{ MHz}$ and the subcarrier spacing is $\frac{1}{KT} = 312.5 \text{ kHz}$. The guard interval is $LT = 0.8 \mu\text{s}$ and therefore channels with impulse responses of this duration can be handled.

[Alternative reading based on visual shape of blurry numbers]:

The chip period T satisfies $(256 + 64)T = 3.2 \mu\text{s}$ and thus $T = 10 \text{ ns}$. Then, $1/T = 100 \text{ MHz}$ and the subcarrier spacing is $\frac{1}{KT} = 390.6 \text{ kHz}$. The guard interval is $LT = 0.64 \mu\text{s}$ and therefore channels with impulse responses of this duration can be handled.

Example 2.23

In LTE, the subcarrier spacing is 15 kHz and every 5 MHz of bandwidth accommodates $K = 300$ subcarriers. What penalty factor should be applied to the spectral efficiency computed with $B = 1/T$?

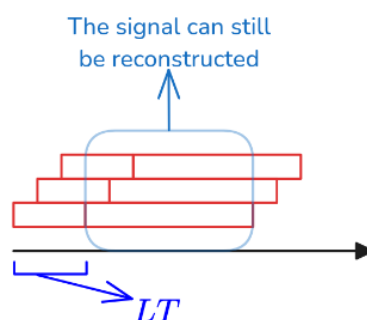
Solution

Since $300 \cdot 15 \cdot 10^3 \text{ Hz}$ out of $5 \cdot 10^6 \text{ Hz}$ are used, the penalty factor equals $\frac{300 \cdot 15 \cdot 10^3}{5 \cdot 10^6} = 0.9$. For NR, this tightens up to 0.96 .

When the solution of the first says *of this duration*, the actual correct thing is *of this duration or less*! In the second problem, the actual amount of carriers would be 333 , but we use less so that we can avoid collision between external carriers.

💀Guard interval

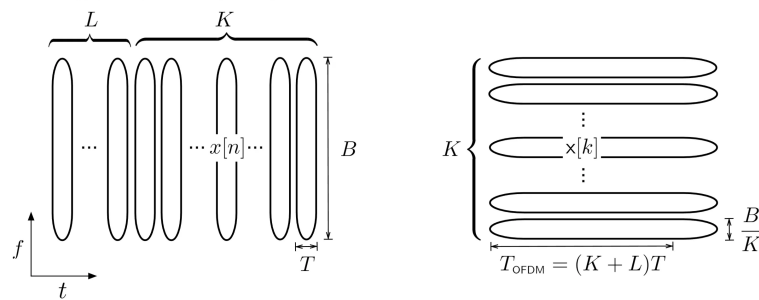
The guard interval is the amount of time occupied by the cyclic prefix; it's called like this because it's an upper bound to how much our signal can be shifted while still being recoverable:



The signal can still be reconstructed (blue arrows indicating LT shift)

 Several definitions: $T, T_{OFDM}, \dots$

Although we never did so in class, it seems necessary to introduce some values, with reference to the following figure (fig. 2.16):



- T is the **chip period**.
- LT is the **guard interval**, as described above.
- T_{OFDM} is the **period of a whole OFDM signal**, given by

$$T_{OFDM} = (K + L)T.$$

- $B = 1/T$ is the **bandwidth** occupied by the **whole amount of subcarriers** collectively.
- $B/K = 1/KT$ is the **bandwidth occupied by a single subcarrier**; it also is the **subcarrier spacing**.

Penalty factor

A completely separate but equally important definition is that of **penalty factor**: it is given by how much the usable bandwidth is in relation to the used bandwidth; the point is that in reality we'll always have some excess bandwidth which will go over our constraints: just to give an example, in *example 2.23* the subcarrier spacing is 15kHz but it turns out that the actual subcarrier bandwidth is $\frac{500\text{MHz}}{300} = 16.66\text{kHz}$, more than what would be available. The penalty factor is thus calculated as

$$\text{Penalty factor} = \frac{\text{Usable bandwidth}}{\text{Used bandwidth}};$$

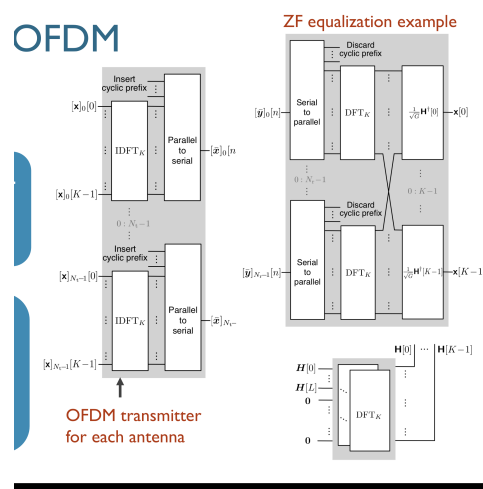
this value is useful as whenever we have a channel of known spectral efficiency R/B but we also know that we're exceeding the channel's bandwidth, we can correct the spectral efficiency as

$$(\text{Penalty factor}) \cdot \frac{R}{B}.$$

Just as an extra piece of information, another way to see this is

$$\text{Penalty factor} = \frac{\text{Theoretically used bandwidth}}{\text{Actually used bandwidth}}.$$

The same structure is found when extending this to MIMO-OFDM:



All the IDFTs at the inputs of the different antennas are of the same sizes; we need the responses to be of the same sizes; L must be the greatest of all channels and K must be calculated knowing the penalty.

If we want to use as few resources as possible, we should customize L, K for every antenna; we won't actually do this.